

MediaPipe pose detection demo

Introduction:

MediaPipe Pose Detection refers to the computer vision approach used for detecting and tracking human body landmarks by using the webcam or video input. The technology is popular in such fields as fitness, motion tracking, virtual try-on, games, and artificial intelligence-based health monitoring systems.

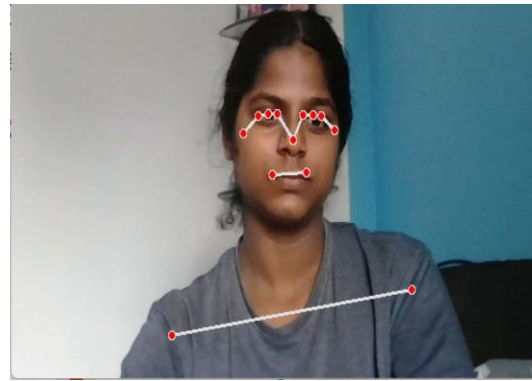
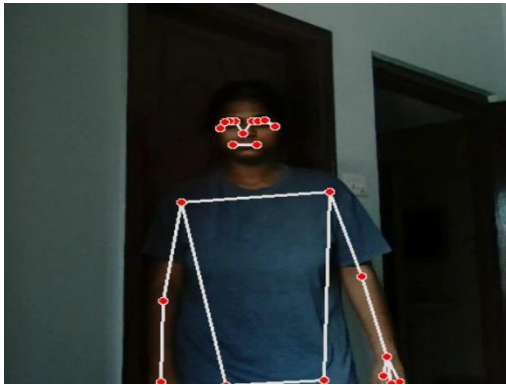
This particular task involved the use of programming languages, such as Python and OpenCV, as well as the MediaPipe library. In order to complete the task, the webcam was used to record live video frames that were further processed by using the MediaPipe library for detecting human body keypoints and connections. As a result, live human body landmarks were shown on the display.

Code:

```
posedetetection.py > ...
1  import cv2
2  import mediapipe as mp
3  mp_pose = mp.solutions.pose
4  pose = mp_pose.Pose()
5  mp_draw = mp.solutions.drawing_utils
6  cap = cv2.VideoCapture(0)
7  while True:
8      success, img = cap.read()
9      if not success:
10         break
11     imgRGB = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
12     results = pose.process(imgRGB)
13     if results.pose_landmarks:
14         mp_draw.draw_landmarks(
15             img,
16             results.pose_landmarks,
17             mp_pose.POSE_CONNECTIONS
18         )
19     cv2.imshow("Pose Detection", img)
20     if cv2.waitKey(1) & 0xFF == ord('q'):
21         break
22 cap.release()
23 cv2.destroyAllWindows()
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
• (venv) PS C:\Users\priyanka\Downloads\raritone internship> python posedetetection.py
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
• (venv) PS C:\Users\priyanka\Downloads\raritone internship> 
```

Output:



Conclusion

In conclusion, the MediaPipe pose detection task was successfully completed based on the use of programming languages Python and OpenCV. In particular, it was possible to use the webcam to record the live video and identify the landmarks of human body in real time. Such a task contributed greatly to my better understanding of computer vision, image processing, and pose estimation.